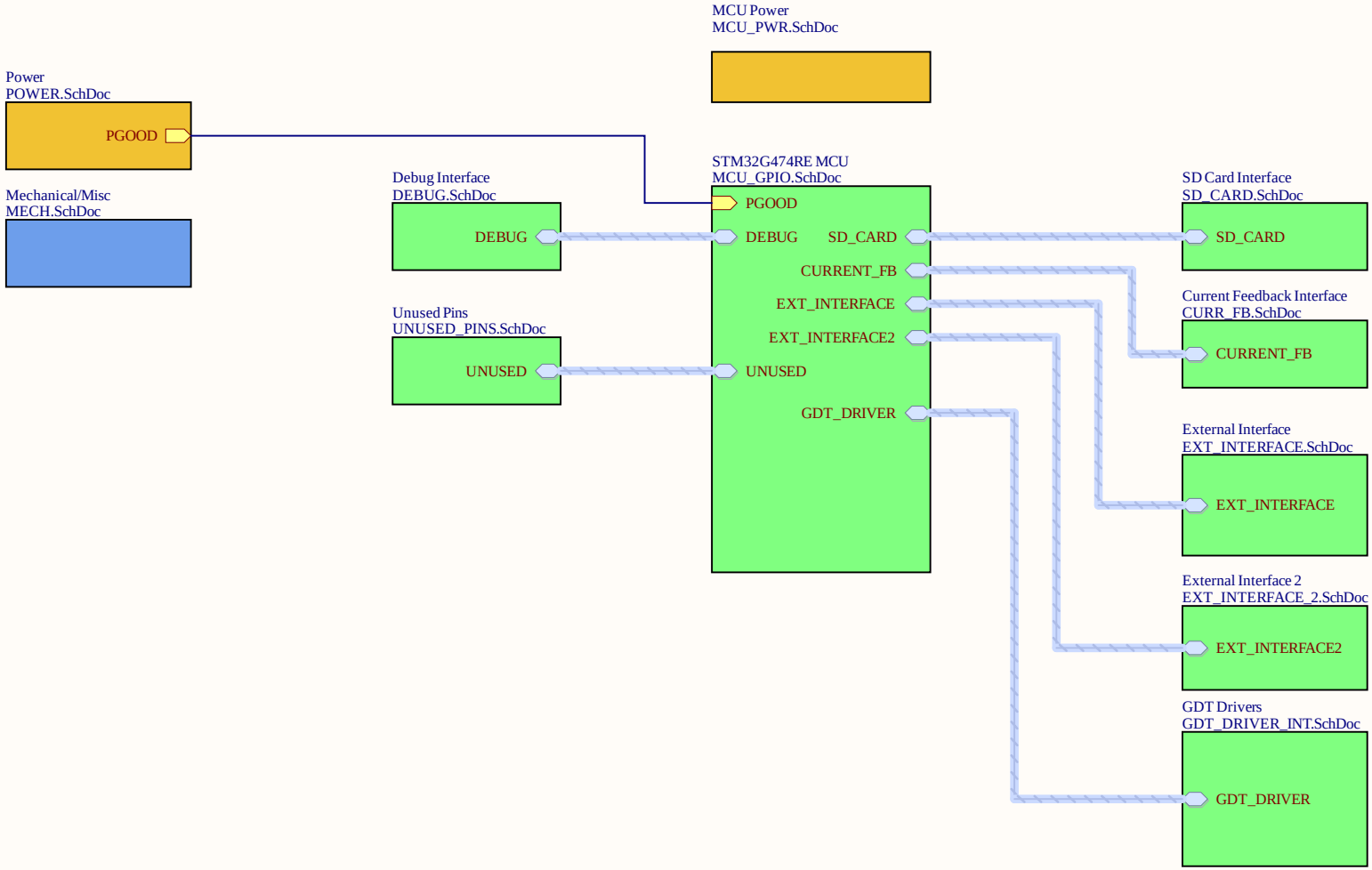
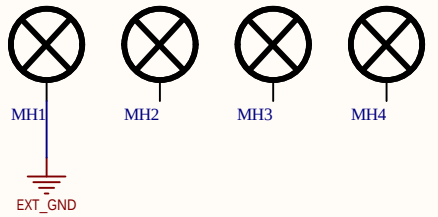


Title: Emma Mk1 DRSSTC Motherboard		github.com/muJoseph	
Description: System Overview			
File: SYSTEM.SchDoc			Rev: Alpha
Size:Letter	Date: 7/27/2026	EE: Joe Corteo	Sheet 1 of 12

Mounting holes



Fiducials



JLCPCB Standard 4 Layer Stack Up

Thickness: 0.8mm | 1.0mm | 1.2mm | **1.6mm** | 2.0mm

Outer Copper Weight: **1oz** | 2oz

inner Copper Weight: **0.5oz** | 1oz | 2oz

1) No requirement Stackup

Layer	Material Type	Thickness	
Top Layer	Copper	0.035mm	
Prepreg	7628*1	0.21040mm	
Inner Layer L2	Copper	0.0152mm	1.1mm H/HOZ with copper
Core>	Core	1.065mm	
Inner Layer L3	Copper	0.0152mm	
Prepreg	7628*1	0.21040mm	
Bottom Layer	Copper	0.035mm	

2) JLC04161H-7628 Stackup

Layer	Material Type	Thickness	
Top Layer	Copper	0.035mm	
Prepreg	7628*1	0.21040mm	
Inner Layer L2	Copper	0.0152mm	1.1mm H/HOZ with copper
Core>	Core	1.065mm	
Inner Layer L3	Copper	0.0152mm	
Prepreg	7628*1	0.21040mm	
Bottom Layer	Copper	0.035mm	

Controlled Impedance PCB Parameters and Stackup

1.Prepreg dielectric constant:

Prepreg type	Dielectric constant
7628	4.4
3313	4.1
1080	3.91
2116	4.16

2.Solder mask Parameters

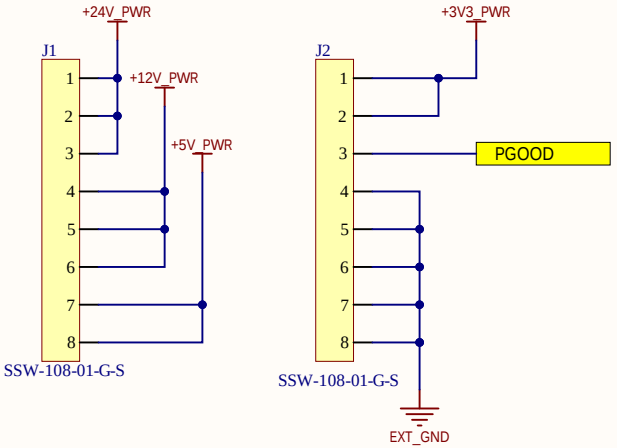
Coating Above Substrate C1	Coating Above Trace C2	Coating Between Traces C3	Coating Dielectric CEr
1.2mil	0.6mil	1.2mil	3.8

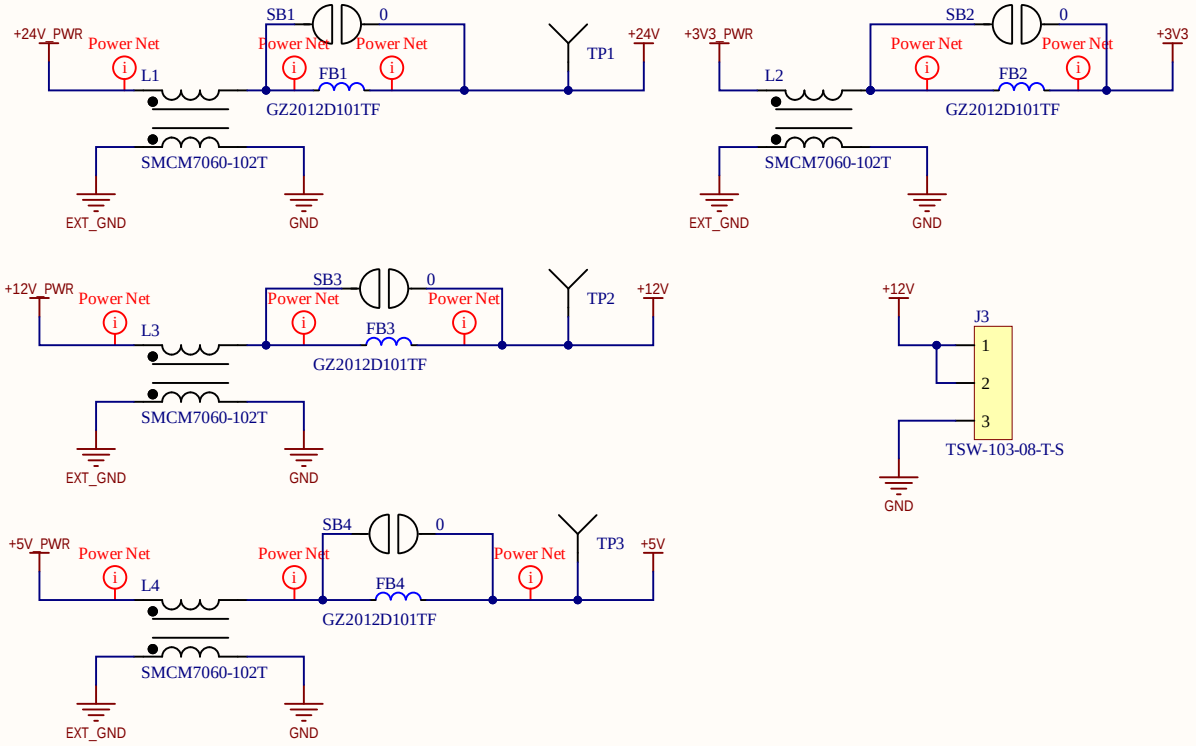
3. Core dielectric constant

Core dielectric constant
4.6

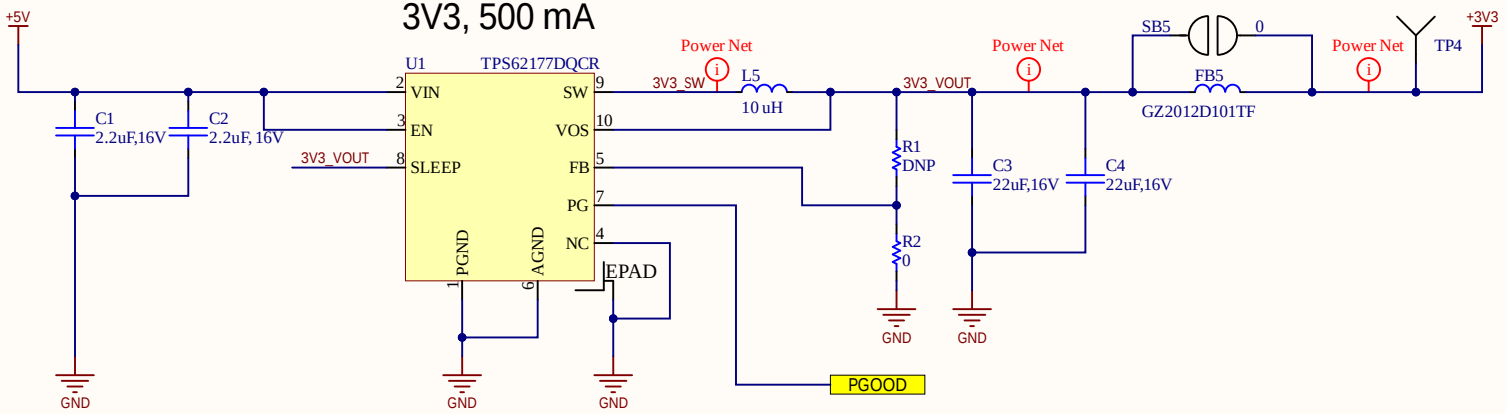
Title: Emma Mk1 DRSSTC Motherboard		github.com/muJoseph	
Description: Mechanical/Misc			
File: MECH.SchDoc			Rev: Alpha
Size:Letter	Date: 7/27/2026	EE: Joe Corteo	Sheet 2 of 12

Power Card Interface



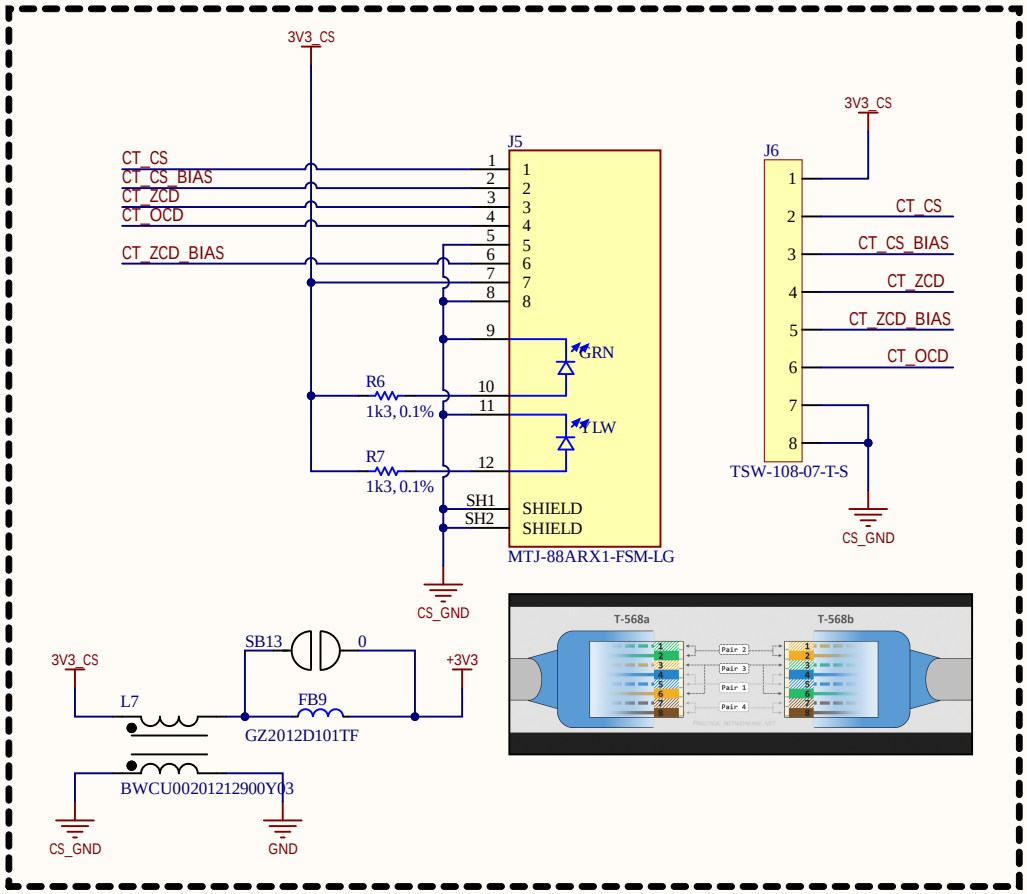


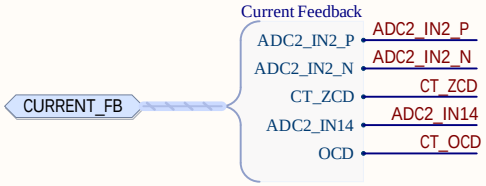
3V3, 500 mA



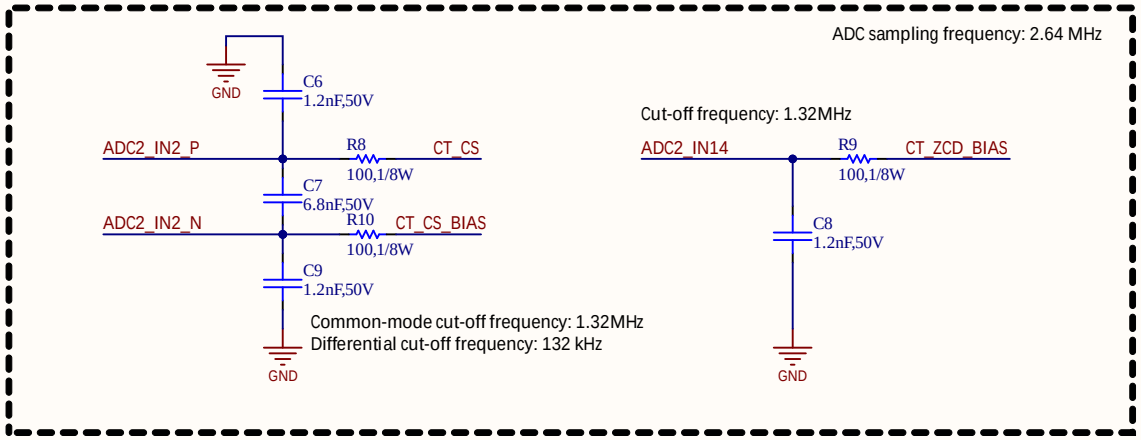
Title: Emma Mk1 DRSSTC Motherboard		github.com/muJoseph	
Description: Power			
File: POWER.SchDoc			Rev: Alpha
Size:Letter	Date: 7/27/2026	EE: Joe Corteo	Sheet 3 of 12

Emma Mk1 DRSSTC CS Interface



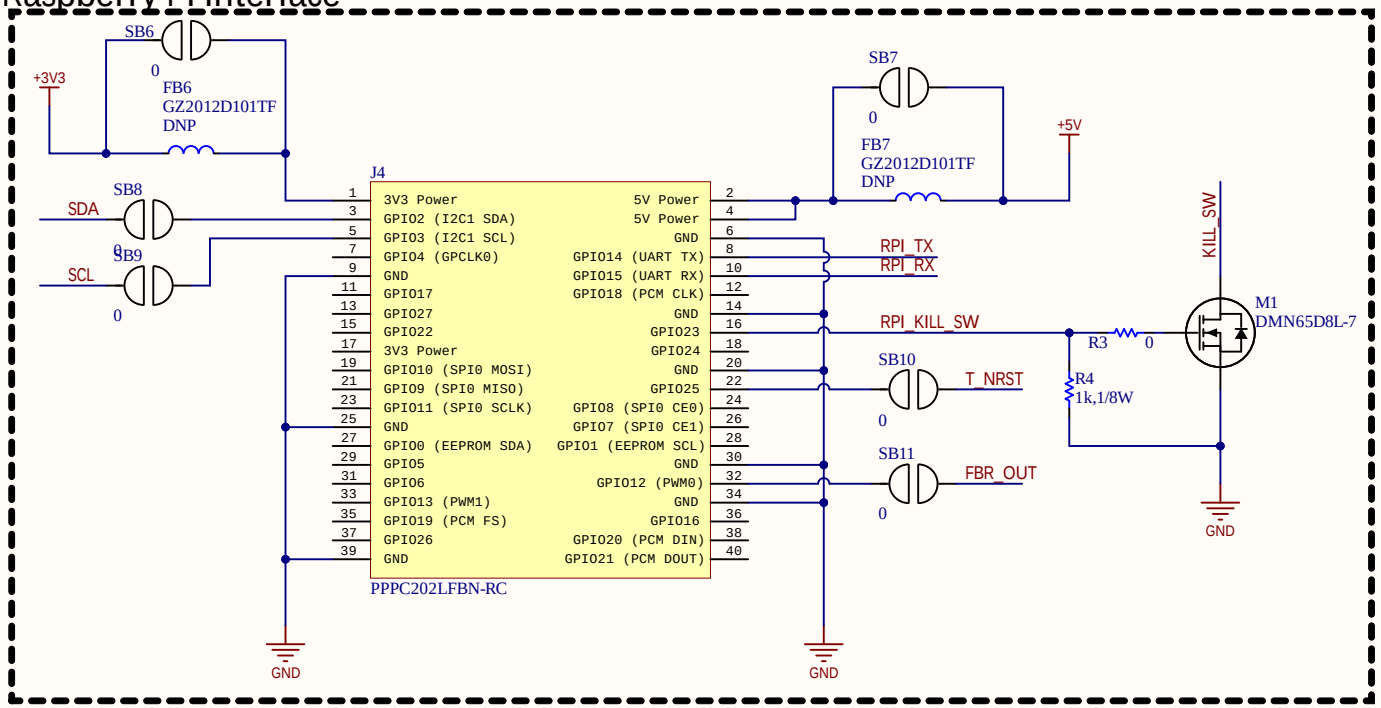


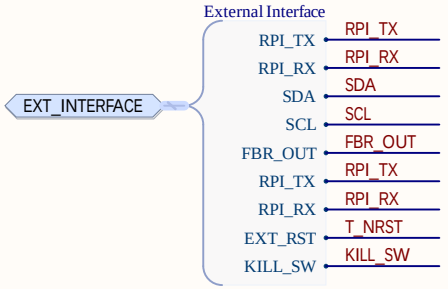
Anti-aliasing filter



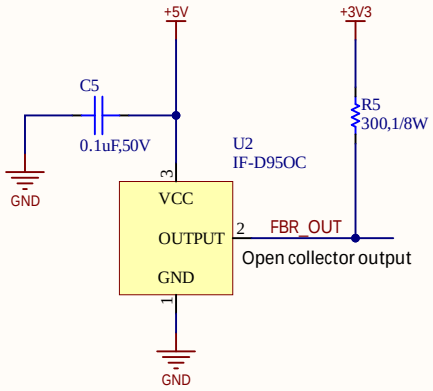
Title: Emma Mk1 DRSSTC Motherboard		github.com/muJoseph	
Description: Current Feedback			
File: CURR_FB.SchDoc			Rev: Alpha
Size: Letter	Date: 7/27/2026	EE: Joe Corteo	Sheet 4 of 12

Raspberry Pi Interface

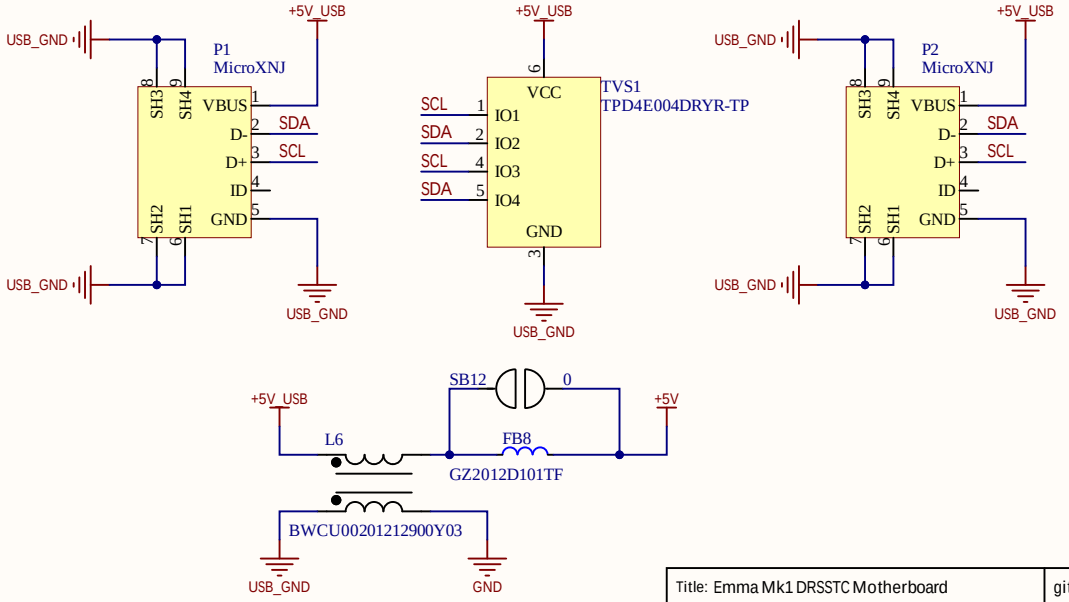




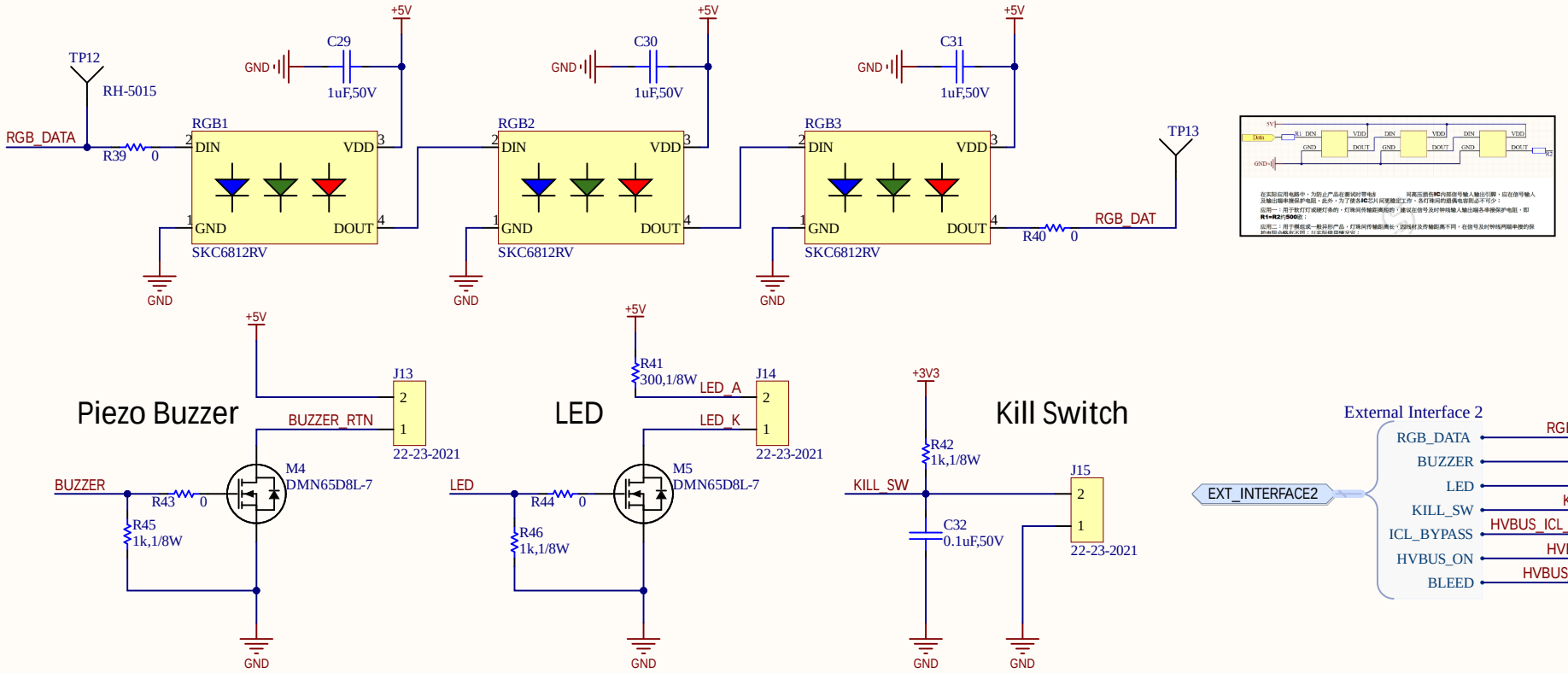
Fiber Optic Receiver



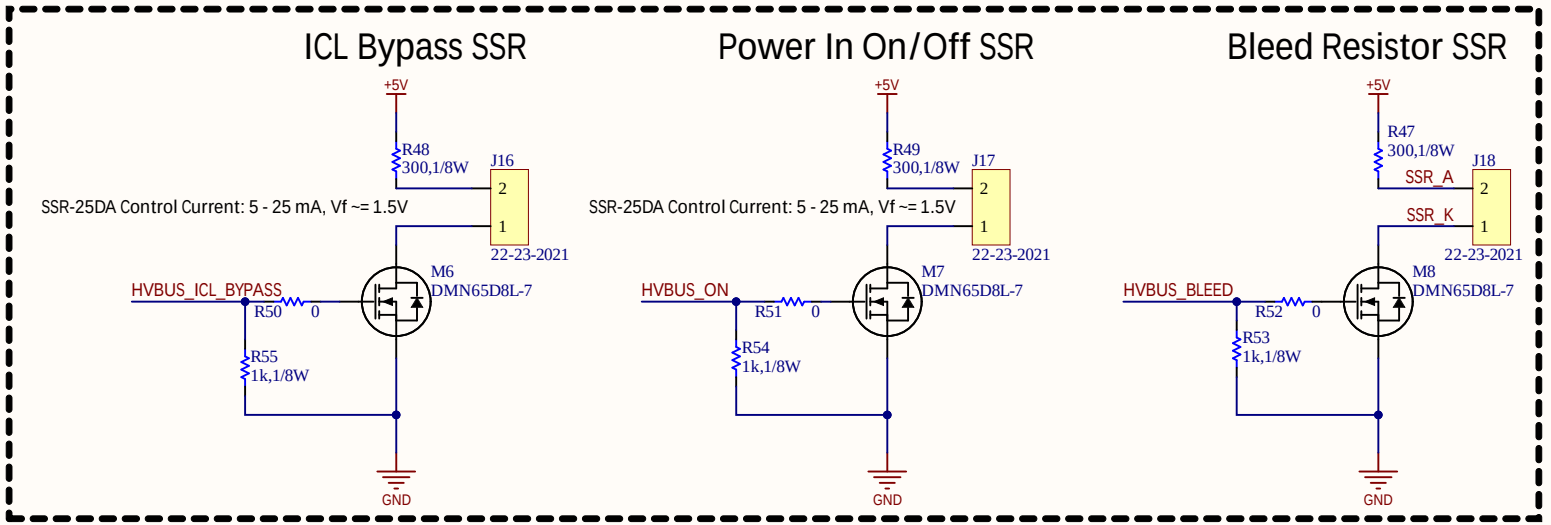
External I2C Bus



Title: Emma Mk1 DRSSTC Motherboard		github.com/muJoseph	
Description: Raspberry Pi and External Sensor Interface			
File: EXT_INTERFACE.SchDoc			Rev: Alpha
Size:Letter	Date: 7/27/2026	EE: Joe Corteo	Sheet 5 of 12



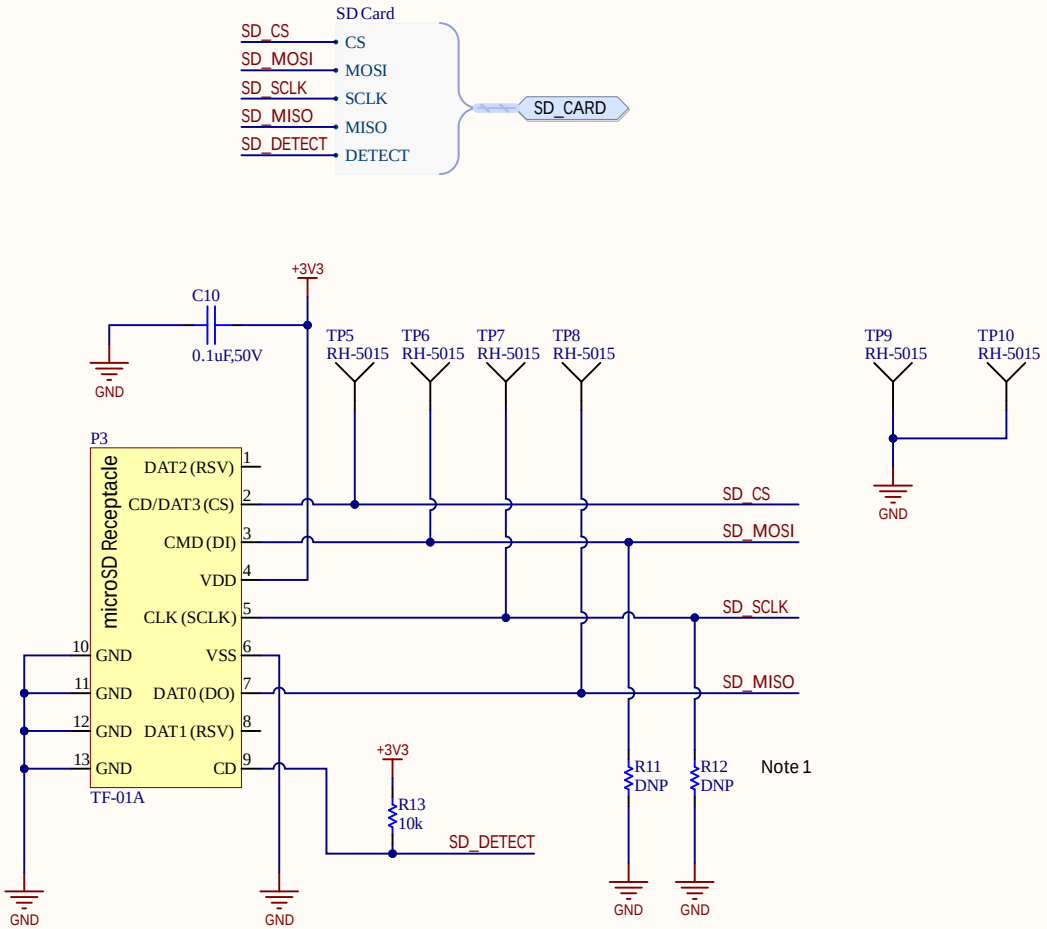
HV Bus Control



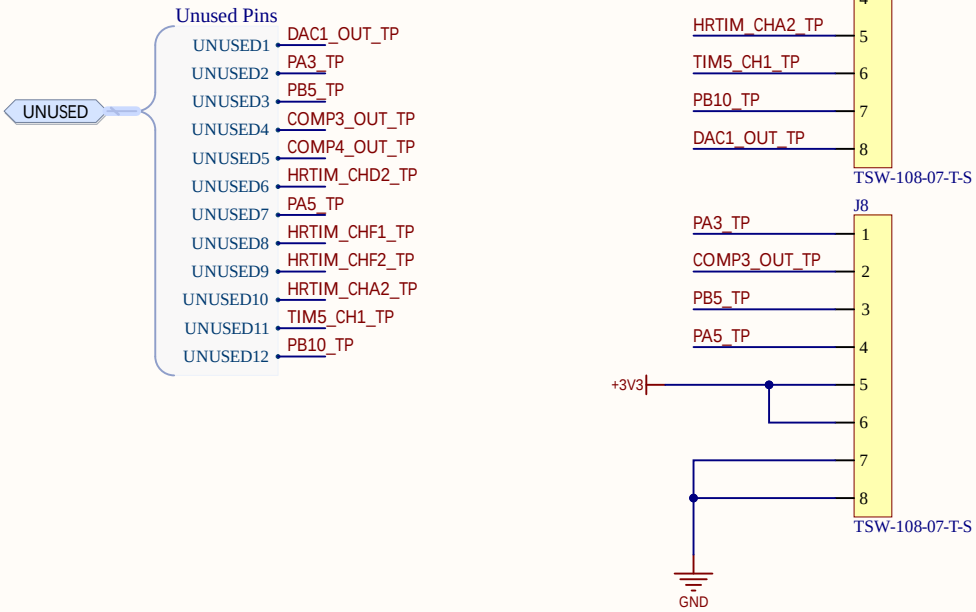
Title: Emma Mk1 DRSSTC Motherboard		github.com/muJoseph	
Description: External Intferface (cont.'d)			
File: EXT_INTERFACE_2.SchDoc			Rev: Alpha
Size:Letter	Date: 7/27/2026	EE: Joe Corteo	Sheet 6 of 12

NOTES

1. Place SPI termination resistors as close to respective destination as possible

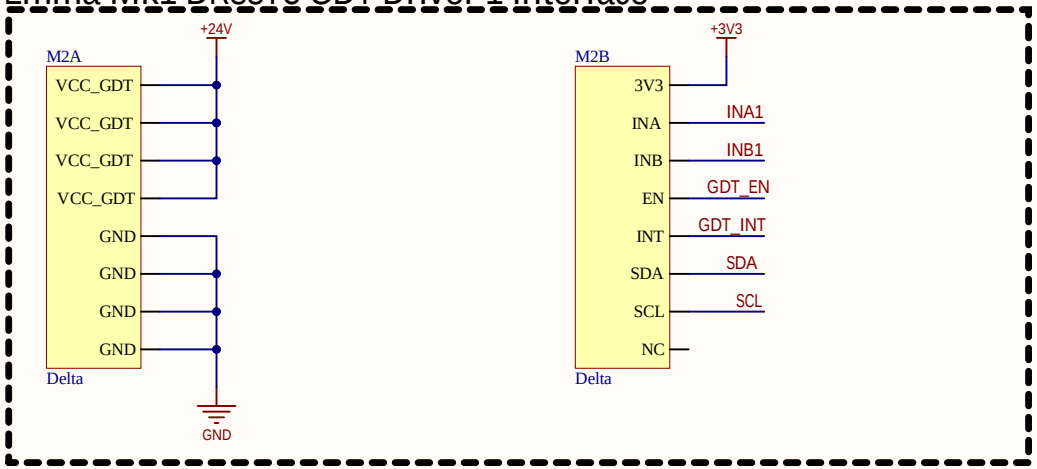


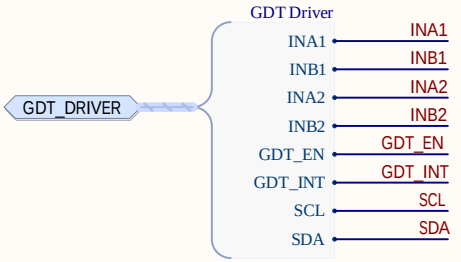
Title: Emma Mk1 DRSSTC Motherboard		github.com/muJoseph	
Description: SD Card Interface			
File: SD_CARD.SchDoc			Rev: Alpha
Size:Letter	Date: 7/27/2026	EE: Joe Corteo	Sheet 7 of 12



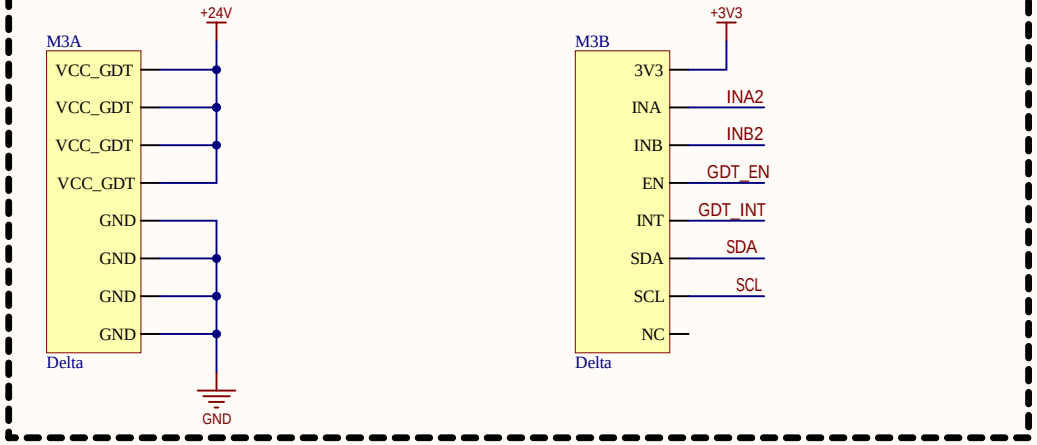
Title: Emma Mk1 DRSSTC Motherboard		github.com/muJoseph	
Description: Unused Pins			
File: UNUSED_PINS.SchDoc			Rev: Alpha
Size:Letter	Date: 7/27/2026	EE: Joe Corteo	Sheet 8 of 12

Emma Mk1 DRSSTC GDT Driver 1 Interface





Emma Mk1 DRSSTC GDT Driver 2 Interface

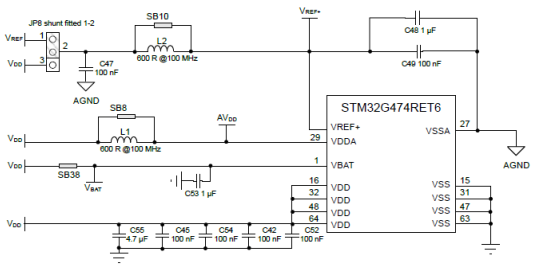


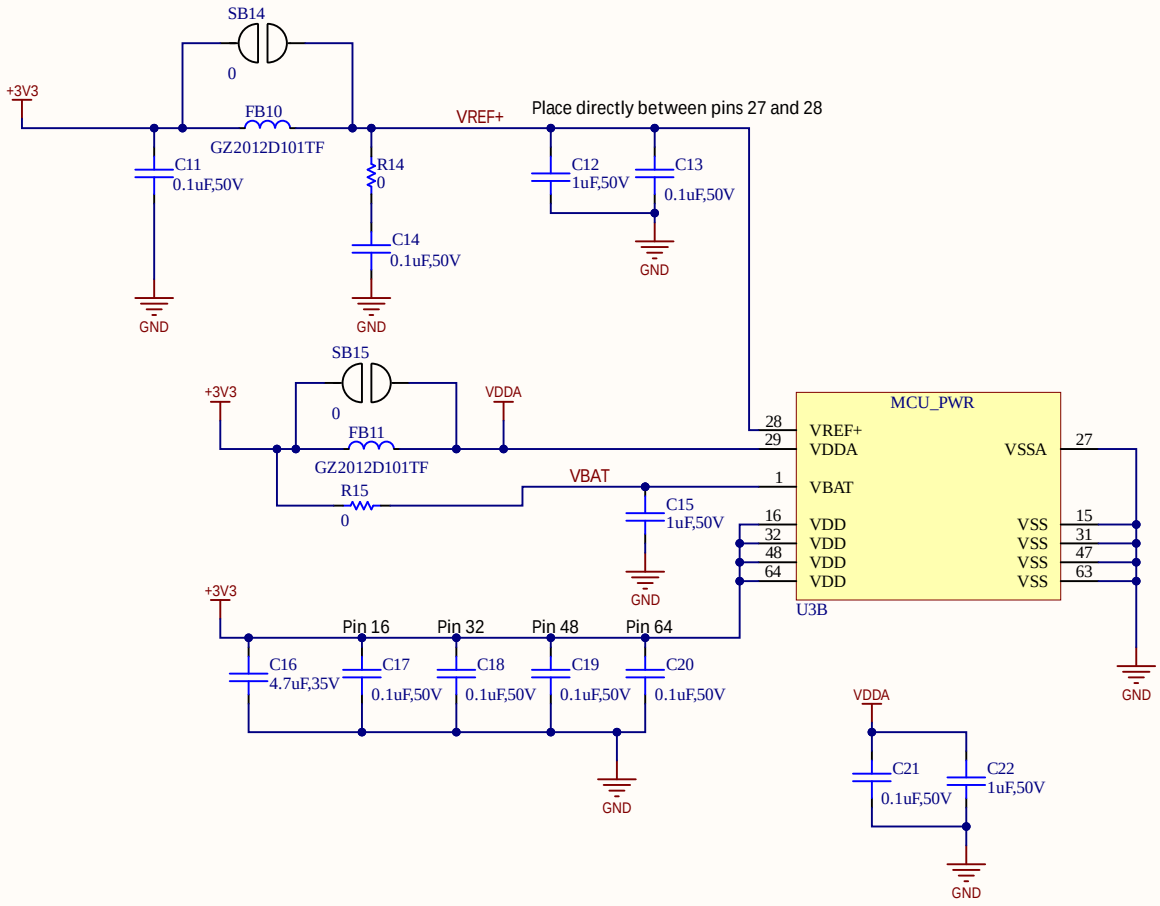
Title: Emma Mk1 DRSSTC Motherboard		github.com/muJoseph	
Description: GDT Driver Interface			
File: GDT_DRIVER_INT.SchDoc			Rev: Alpha
Size:Letter	Date: 7/27/2026	EE: Joe Corteo	Sheet 9 of 12

Filtering of analog supply voltages:

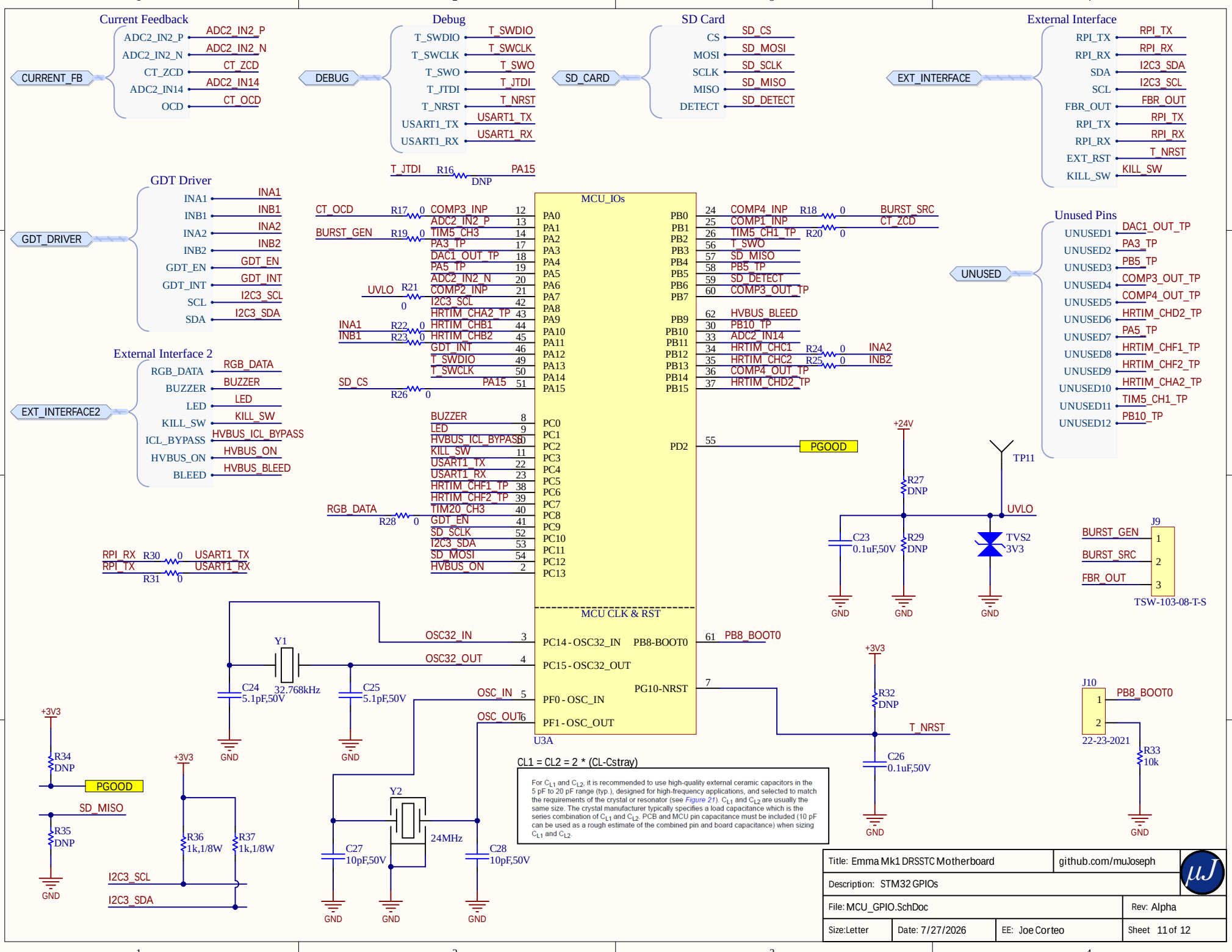
- Avoid to use of larger inductors as filters for VSSA or VREF+ pins. The inductor together with the decoupling capacitor create an LC filter that operates correctly only if its load is stable (like some resistance on the LC filter output). It filters correctly the input voltage fluctuations. If the LC filter load generates current peaks (like current peaks from VREF+ or VDDA pin), this fast transient current peak can lead to attenuated oscillations of this LC filter at its resonant frequency. In fact, the current peak decreases the voltage on the decoupling capacitor and, because the inductor inhibits large current changes, the capacitor is charged back slowly and the LC circuit may oscillate. These voltage oscillations are present on the decoupling capacitor and therefore, the V_{REF+} or V_{DDA} voltage is influenced. The oscillations on VREF+ or VDDA pin are attenuated after a relative long time (when the current peak disappears) and may influence the V_{REF+} and V_{DDA} voltages in later critical times.
- In some designs, the high frequency inductor (or better say choke) must be used to filter very high noise from the power supply (if really present). In parallel to this choke footprint, it is recommended to design the solder bridge that can short the choke in case of problems with V_{REF+} or V_{DDA} voltage stability (see the example in the figure below). The ferrite choke must be properly selected to filter very high frequencies (relative to current supply peaks frequency) and their impedance character at high frequencies must be more resistive than inductive (high-frequency losses in the ferrite core).

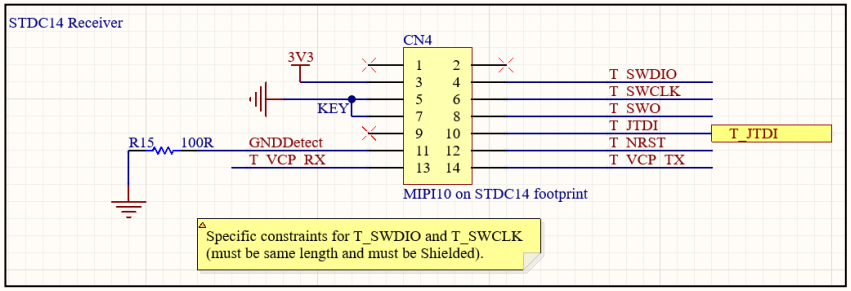
Figure 5. Example of designing solder bridges (SB8, SB10) in parallel to chokes (L1, L2) on analog supply voltages

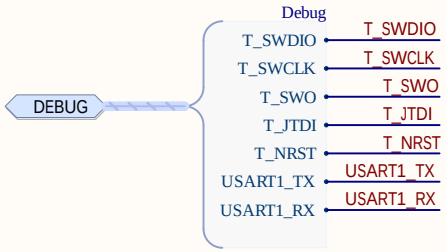


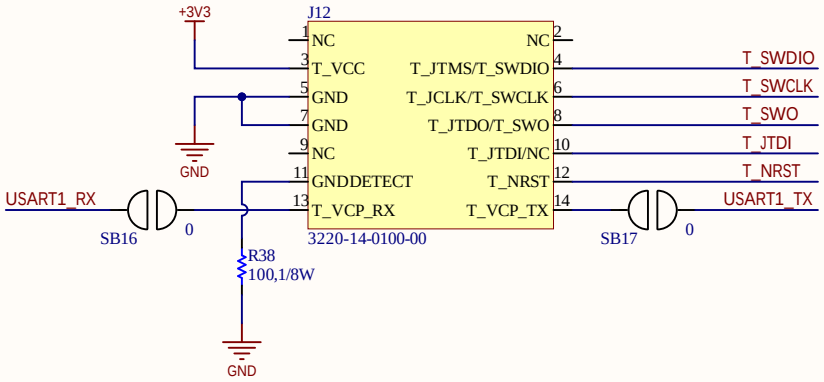


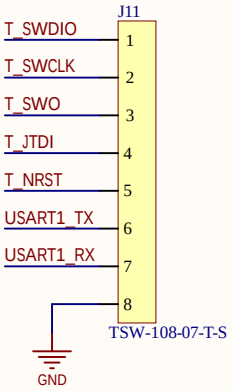
Title: Emma Mk1 DRSSTC Motherboard		github.com/muJoseph	
Description: STM32 Power			
File: MCU_PWR.SchDoc			Rev: Alpha
Size:Letter	Date: 7/27/2026	EE: Joe Corteo	Sheet 10 of 12


Current Feedback
CURRENT_FB
ADC2_IN2_P
ADC2_IN2_N
CT_ZCD
ADC2_IN14
OCD
Debug
DEBUG
T_SWDIO
T_SWCLK
T_SWO
T_JTDI
T_NRST
USART1_TX
USART1_RX
SD Card
SD_CARD
CS
MOSI
SCLK
MISO
DETECT
SD_CS
SD_MOSI
SD_SCLK
SD_MISO
SD_DETECT
External Interface
EXT_INTERFACE
RPI_TX
RPI_RX
SDA
SCL
FBR_OUT
EXT_RST
KILL_SW
I2C3_SDA
I2C3_SCL
T_NRST
GDT Driver
GDT_DRIVER
INA1
INB1
INA2
INB2
GDT_EN
GDT_INT
External Interface 2
EXT_INTERFACE2
RGB_DATA
BUZZER
LED
KILL_SW
ICL_BYPASS
HVBUS_ICL_BYPASS
HVBUS_ON
BLEED
HVBUS_BLEED
MCU_IOs
PA0
PA1
PA2
PA3
PA4
PA5
PA6
PA7
PA8
PA9
PA10
PA11
PA12
PA13
PA14
PA15
PB0
PB1
PB2
PB3
PB4
PB5
PB6
PB7
PB9
PB10
PB11
PB12
PB13
PB14
PB15
PC0
PC1
PC2
PC3
PC4
PC5
PC6
PC7
PC8
PC9
PC10
PC11
PC12
PC13
PD2
MCU CLK & RST
PC14 - OSC32_IN
PC15 - OSC32_OUT
PF0 - OSC_IN
PF1 - OSC_OUT
PB8-BOOT0
PG10-NRST
U3A
Unused Pins
UNUSED
UNUSED1
UNUSED2
UNUSED3
UNUSED4
UNUSED5
UNUSED6
UNUSED7
UNUSED8
UNUSED9
UNUSED10
UNUSED11
UNUSED12
DAC1_OUT_TP
PA3_TP
PB5_TP
COMP3_OUT_TP
COMP4_OUT_TP
HRTIM_CHD2_TP
PA5_TP
HRTIM_CHF1_TP
HRTIM_CHF2_TP
HRTIM_CHA2_TP
TIM5_CH1_TP
PB10_TP
T_JTDI
R16
DNP
PA15
CT_OCD
R17
0
COMP3_INP
ADC2_IN2_P
BURST_GEN
R19
TIM5_CH3
PA3_TP
DAC1_OUT_TP
PA5_TP
ADC2_IN2_N
UVLO
R21
COMP2_INP
I2C3_SCL
HRTIM_CHA2_TP
INA1
R22
HRTIM_CHB1
INB1
R23
HRTIM_CHB2
GDT_INT
T_SWDIO
T_SWCLK
SD_CS
R26
BUZZER
LED
HVBUS_ICL_BYPASS
KILL_SW
USART1_TX
USART1_RX
HRTIM_CHF1_TP
HRTIM_CHF2_TP
RGB_DATA
R28
TIM20_CH3
GDT_EN
SD_SCLK
I2C3_SDA
SD_MOSI
HVBUS_ON
RPI_RX
R30
RPI_TX
R31
COMP4_INP
R18
BURST_SRC
CT_ZCD
COMP1_INP
TIM5_CH1_TP
R20
T_SWO
SD_MISO
PB5_TP
SD_DETECT
COMP3_OUT_TP
HVBUS_BLEED
PB10_TP
ADC2_IN14
HRTIM_CHC1
R24
INA2
HRTIM_CHC2
R25
INB2
COMP4_OUT_TP
HRTIM_CHD2_TP
PGOOD
+24V
R27
DNP
TP11
UVLO
C23
0.1uF,50V
R29
DNP
TVS2
3V3
GND
J9
BURST_GEN
BURST_SRC
FBR_OUT
TSW-103-08-T-S
Y1
OSC32_IN
OSC32_OUT
32.768kHz
C24
5.1pF,50V
C25
5.1pF,50V
OSC_IN
OSC_OUT
+3V3
R34
DNP
PGOOD
SD_MISO
R35
DNP
R36
1k,1/8W
R37
1k,1/8W
I2C3_SCL
I2C3_SDA
Y2
24MHz
C27
10pF,50V
C28
10pF,50V
CL1 = CL2 = 2 * (CL-Cstray)
For CL1 and CL2, it is recommended to use high-quality external ceramic capacitors in the 5 pF to 20 pF range (typ.), designed for high-frequency applications, and selected to match the requirements of the crystal or resonator (see Figure 21). CL1 and CL2 are usually the same size. The crystal manufacturer typically specifies a load capacitance which is the series combination of CL1 and CL2. PCB and MCU pin capacitance must be included (10 pF can be used as a rough estimate of the combined pin and board capacitance) when sizing CL1 and CL2.
R32
DNP
T_NRST
C26
0.1uF,50V
J10
PB8_BOOT0
22-23-2021
R33
10k
Title: Emma Mk1 DRSSTC Motherboard
github.com/muJoseph
Description: STM32 GPIOs
File: MCU_GPIO.SchDoc
Rev: Alpha
Size:Letter
Date: 7/27/2026
EE: Joe Corteo
Sheet 11 of 12

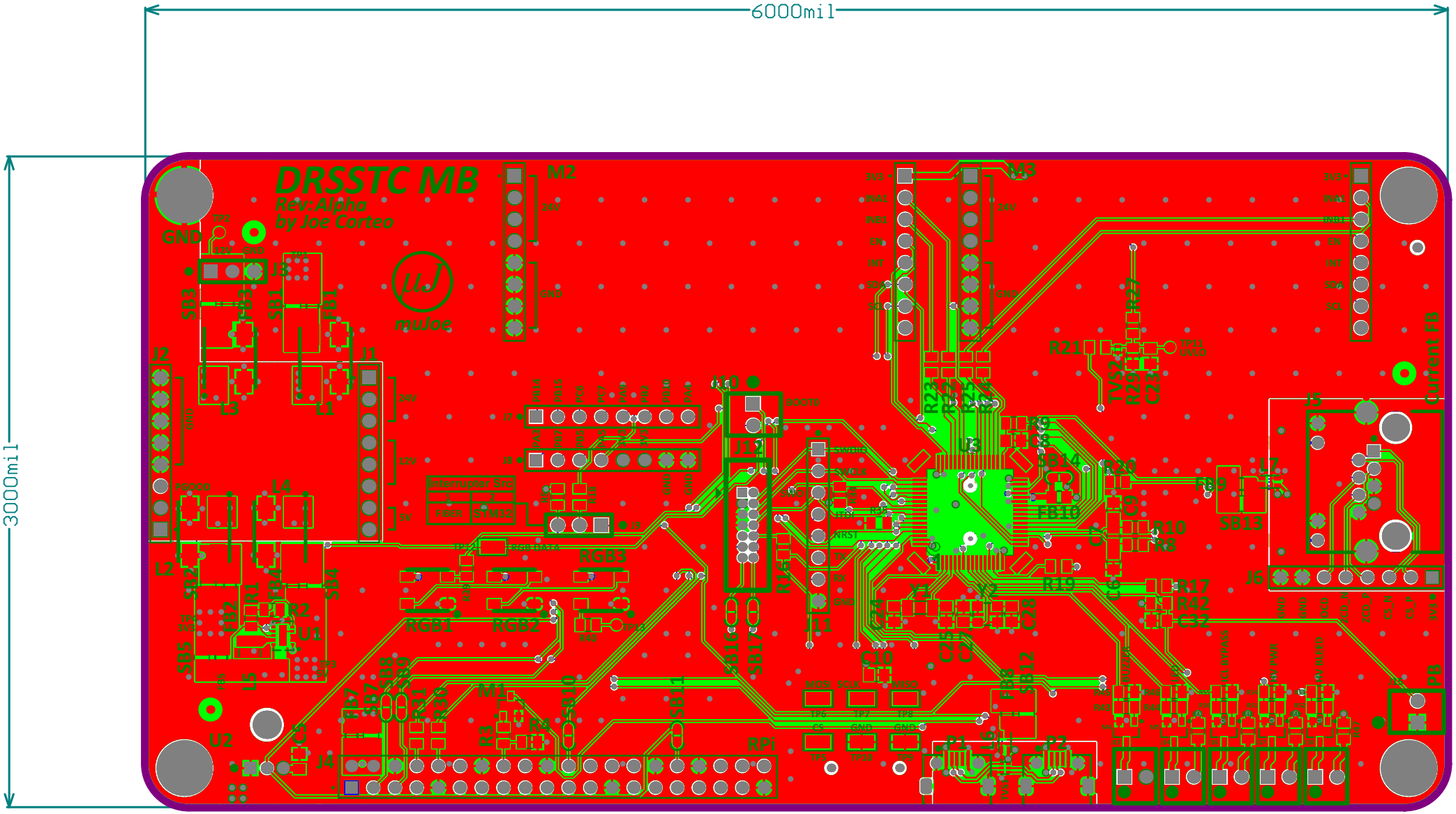








Title: Emma Mk1 DRSSTC Motherboard		github.com/muJoseph	
Description: Debug Interface			
File: DEBUG.SchDoc			Rev: Alpha
Size:Letter	Date: 7/27/2026	EE: Joe Corteo	Sheet 12 of 12


6000mil
3000mil
DRSTC MB
Rev:Alpha
by Joe Corteo
mJoe
GND
12V
24V
3V
PGOOD
Interrupter Src
FIBER
STM32
RGB1
RGB2
RGB3
SWDIO
SWCLK
NRST
BOOT0
MOSI
SCLK
MISO
CS
RPi
Current FB
PB